

MAC-2313 | CALCULUS III

Practice Exam 2

Original paper-style practice • no formula sheet

Name

Date

Score

Eight core problems, 10 points each. One optional bonus problem is worth up to 10 additional points. Show your reasoning: a correct method with a small arithmetic error may earn partial credit.

No formula sheet is provided. Assume the single-variable derivative and integral skills from Calculus I and substitution/integration by parts from Calculus II are available when useful. Exact answers are preferred unless an estimate is requested.

Suggested practice: attempt the exam without notes, then use the separate answer key to diagnose method choices and reasoning. No time limit is specified.

Problem 1 | Vectors and displacement

Show enough work for partial credit. State units and interpret your result when relevant.

An underwater robot travels from $A=(0,2,1)$ to $B=(3,-1,4)$, then to $C=(-1,3,6)$, in meters.

(a) Compute each leg displacement and net displacement. (b) Compare total path length with endpoint distance. (c) Give the net direction as a unit vector. (d) If the robot returns directly from C to A, what is the new displacement and what is the total distance of the closed trip? Explain the difference between a zero net displacement and zero distance.

Problem 2 | Dot products, projection, and work

Show enough work for partial credit. State units and interpret your result when relevant.

A force $\mathbf{F} = \langle -2, 5, 4 \rangle$ N acts during displacement $\mathbf{d} = \langle 2, -1, 2 \rangle$ m.

(a) Compute the work and determine whether the force is mostly aligned or opposed to motion. (b) Find the scalar and vector projections of $\mathbf{F}$ onto $\mathbf{d}$. (c) Find $\cos(\theta)$ and classify the angle. (d) Explain how negative work can occur even though the force is nonzero.

Problem 3 | Cross products and plane equations

Show enough work for partial credit. State units and interpret your result when relevant.

A laser sheet is intended to pass through $A=(0,1,2)$, $B=(2,0,1)$, and $C=(1,3,0)$. A sensor at $D=(1,1,1)$ may be in the sheet.

(a) Construct a normal and plane equation. (b) Find both the parallelogram and triangle areas determined by the three markers. (c) Test whether D lies in the plane. (d) Explain how you would detect if the first three markers were collinear using the same method.

Problem 4 | First and second partial derivatives

Show enough work for partial credit. State units and interpret your result when relevant.

Let $f(s,t)=e^s \sin(t)+s t^2$ describe a response surface. At $(0,\pi/2)$, find f_s , f_t , f_{ss} , f_{st} , f_{ts} , and f_{tt} . State the estimated first-order response change for $(ds,dt)=(0.01,-0.02)$.

Problem 5 | Differentials and approximation

Show enough work for partial credit. State units and interpret your result when relevant.

A cylindrical tank's full outside surface area is $S(r,h)=2\pi r^2+2\pi rh$. At $r=4$ cm and $h=7$ cm, a redesign changes $dr=-0.03$ cm and $dh=0.04$ cm.

(a) Estimate dS and its sign. (b) Estimate the relative percent change. (c) Explain which dimensional change dominates, and identify why multiplying the separate percentage changes would not correctly estimate the area change.

Problem 6 | Multivariable chain rule

Show enough work for partial credit. State units and interpret your result when relevant.

Let $T(x,y,z)=x^2y+yz$, where $x=t^2+1$, $y=3-t$, and $z=2t$. Find dT/dt at $t=1$ by the multivariable chain rule. Then substitute first to obtain a single-variable expression and verify the result.

Problem 7 | Gradient and directional derivative

Show enough work for partial credit. State units and interpret your result when relevant.

A cost field is $c(x,y)=3x^2+y^2-2xy$. At $P=(1,1)$, evaluate how cost changes toward $Q=(-2,5)$. Also find the direction and rate of fastest local increase and the direction of fastest decrease.

Problem 8 | Constrained optimization

Show enough work for partial credit. State units and interpret your result when relevant.

A workshop's profit model is $P(x,y)=22x+35y-x^2-2xy-3y^2$, with production $x,y\geq 0$ and capacity $x+y\leq 10$. Units of P are hundreds of dollars.

(a) Find the unconstrained interior critical point and determine whether it is feasible. (b) Optimize P on the capacity edge $x+y=10$. (c) Check the other edges and vertices sufficiently to identify the constrained maximum. Explain why the unconstrained answer cannot simply be accepted.

Optional bonus — variable-force work | 10 optional points

Show enough work for partial credit. State units and interpret your result when relevant.

Along the x -axis from $x=0$ to $x=\pi/2$ m, a particle experiences force $F(x)= N$. Find the work using substitution. Explain why the displacement vector alone does not determine work here.
